

Bhumika Agarwal

bhumika.aga@gmail.com — [LinkedIn](#) — [GitHub](#) — [Portfolio](#)

Software Engineer with **4 years** of experience designing scalable backend systems using **Java, Spring Boot, Microservices, Camunda 7 (BPMN), PostgreSQL, and AWS**. Skilled in modernizing legacy platforms, designing distributed systems, and optimizing API performance, delivering **10K+ dispute cases per month**.

Experience

JP Morgan Chase & Co.

Software Engineer II

Oct 2024 – Present

Mumbai, Maharashtra

- Own the architecture and delivery of the **Overcharge Dispute** capability, building Java services on **Spring Boot + Camunda 7** that process **10K+ dispute cases/month**.
- Modernized legacy **Pega** workflows into **10+ reusable Camunda BPMN components** implementing business rules, retries, exception handling, and orchestration, **reducing migration effort by 40%**.
- Automated **AWS** infrastructure provisioning across **3+ environments** using **Terraform (IaC)**, **reducing deployment time by 60%** and improving deployment consistency.
- Established reusable context and prompt conventions for **AI-assisted development** (Claude, GitHub Copilot), standardizing engineering workflows and accelerating legacy modernization.
- Resolved **50+ production issues**, supported **20+ production releases**, and improved service reliability through automated testing.

Cognizant Technology Solutions

Software Engineer

Jul 2022 – Jul 2024

Mumbai, Maharashtra

- Developed **5+ Spring Boot microservices** and **40+ REST APIs** supporting scalable BFSI transaction platforms.
- Optimized Hibernate/JPA and SQL queries across **20+ endpoints**, **reducing average API latency by 30%**.
- Expanded automated testing with **JUnit 5 and Mockito**, improving release quality and reducing regression defects.
- Performed **150+ code reviews**, production deployments, RCA, and cross-team technical collaboration.

Projects

Regulatory Approval System | Java 17, Spring Boot 3, Camunda 7, Spring Security, JWT, Docker

[Live Demo](#)

- Engineered a BPMN workflow engine using **5 Camunda workers** with **8-48 hr SLA**, retries, orchestration, and failure recovery.
- Implemented Spring Security, JWT, and **6-role RBAC**, securing **25+ REST APIs** across approval workflows.

MediFlow | Java, Spring Boot, Microservices, JWT, PostgreSQL, Docker

[Live Demo](#)

- Built a mail-order pharmacy platform on **5 independent Spring Boot microservices** (auth, drugs, refill, subscription, and a Swagger/OpenAPI aggregator), each loosely coupled with a **database-per-service** design.
- Implemented **stateless JWT (HS512)** authentication validated independently by each service (no central runtime auth dependency), and containerized all services with **Docker Compose**.

Education

University of Petroleum and Energy Studies (UPES)

B.Tech. in Computer Science and Engineering

Jun 2018 – Jun 2022

Dehradun, Uttarakhand

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

Skills

Languages & Databases: Java, SQL, JavaScript, TypeScript, PostgreSQL, MySQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, Microservices, REST APIs, Camunda 7

Cloud & DevOps: AWS, Docker, Terraform (IaC), Maven, Jenkins, CI/CD

Testing & Concepts: JUnit, Mockito, System Design, Distributed Systems, AI-Assisted Development (Claude, Copilot)